November 2 — final contact scores complete

Both jobs succeeded around 14:08 EST. Each scored 97 eval-val proteins and 19 eval-denovo designs with 100 rollouts per protein. All 23,200 rollouts finished; zero extensions, failures or preemptions. No experiment training or evaluation jobs remain active.

All-range R-precision, natural / designed: two-epoch 5e-4 0.52470 / 0.67385; two-epoch 1e-3 0.47779 / 0.66229; one-epoch GPU 0.46301 / 0.66515; one-epoch TPU 0.46018 / 0.66954; full-data default 0.55510 / 0.69640.

The final 5e-4 model gains 0.06170 on eval-val over its one-epoch fork, remaining 0.03040 below the default. Training exposure and schedules differ. Designs number only 19; matched sequence-KNN null is unavailable.

Per-protein metrics, timings, bootstrap intervals, provenance and comparisons are committed and public. October 29 results remain archived. The default remains unchanged. At the user's request, the issue 373 agent will update that study to the final two-epoch 5e-4 model; exact checkpoint and artifact details are in the README handoff.

November 2 — two-epoch training complete

The 1e-3 parent finished October 31 at 13:25 EDT; 5e-4 finished November 1 at 15:33 EST. Both stopped at checkpoint 373768 after exactly 373769 optimizer updates. No training jobs remain active; 2e-3 remains cancelled.

Final full LM validation loss: 5e-4 two epochs 3.01329; 1e-3 two epochs 3.04157. Completed one-epoch forks: GPU 3.03572, TPU 3.03582. Full-data default: 2.98441.

The two-epoch 5e-4 model leads the redesign-only runs, improving by 0.02243 over its one-epoch fork, but remains 0.02888 above the full-data default. Different exposures and schedules limit causal interpretation.

Regional CPU verification read every exported parameter: all are finite. Native trainer, optimizer, LR-schedule and inner-optimizer counters agree; both tokenizers are co-located. Verification manifests and complete validation curves are saved.

Final two-epoch R-precision is complete and shown in the preceding slide. October 29 parent contact scores describe earlier checkpoints. The optional published one-epoch model and exp277 default remain unchanged; the separate diversity/Neff study is issue 373.

October 30 — one-epoch GPU checkpoint published

Selected contacts-v1-exp364-mpnn-only-1ep-lr5e-4-cooldown-1.5B, step 186884, for a proposed eval-val diversity study. Eval-val R-precision: GPU 0.46301 versus TPU 0.46018. This is a practical tie; selection uses the relevant point estimate, not a claim of GPU superiority.

All six evaluated model/tokenizer files (5.89 GB) are published unchanged in the public HF bucket. A regional CoreWeave CPU job performed the transfer. Anonymous object sizes and Xet hashes match; small files passed SHA-256 readback and config/tokenizer loading checks. Publication source, launch and verification manifests are committed.

MODELS.yaml registers the checkpoint as optional. Exp277 remains the default; 14 registry tests pass. A separate diversity issue is awaiting the user's review before filing. No diversity analysis has run. Existing training jobs continue independently.

October 30 — 2e-3 stopped at the user's request

The unstable 2e-3 trial was cancelled at 15:44:10 UTC (11:44 EDT). The root and all eight GPU worker tasks are verified killed, releasing 64 H100s. Last logged step: 255772, approximately 1.369 epochs; latest validation loss: 3.67563.

The 5e-4 and 1e-3 parents continue unchanged on 32 H100s each and both made new optimizer progress after the stop. Their existing two-epoch targets remain in force. Both one-epoch forks are complete.

Keep the cancelled trial's checkpoints, metrics and October 29 evaluation artifacts. It is excluded from future recovery unless explicitly reauthorized; it did not complete two epochs. Contact scores are unchanged.

Earlier October 30 — lower learning rates are cooling down

At 15:30 UTC (11:30 EDT), all three two-epoch runs are advancing on 128 H100s. Epochs completed for 5e-4 / 1e-3 / 2e-3: 1.666 / 1.842 / 1.366. Latest validation: 3.07204 / 3.08277 / 3.67563.

Both lower-LR runs are in cooldown and improving. The 2e-3 run remains unstable: a new validation spike reached 4.21406, followed by partial recovery and another rise. Its requested LR is preserved.

Recent elapsed-time pace suggests 1e-3 finishes October 31 afternoon and the other two November 1 evening, Eastern. This includes observed overhead; future preemptions can delay completion. Active compute remaining is approximately 48 / 23 / 50 hours.

Both one-epoch forks remain complete. The five-checkpoint October 29 R-precision snapshot is unchanged; these are not scores for today's later parent checkpoints. No training dispatch or configuration was changed.

October 29 — both cooldown forks are complete

GPU final validation: 3.03572; TPU final validation: 3.03582. Both have exact native counters of 186885 updates at checkpoint 186884 and verified finite exported weights.

At 23:03 UTC, the 5e-4 / 1e-3 / 2e-3 parent runs are at 1.567 / 1.740 / 1.175 epochs of two. Latest validation losses: 3.07851 / 3.09601 / 3.34287. Active compute remaining: approximately 62 / 37 / 65 hours, plus interruptions and validation/checkpoint overhead.

The high-LR run remains unstable. Its requested learning rate is preserved.

October 29 — new contact evaluation

Checkpoint steps: 282830 (5e-4), 314987 (1e-3), 207662 (2e-3), and final 186884 for both cooldown forks.

All-range R-precision, natural / designed: GPU cooldown 0.4630 / 0.6652; TPU cooldown 0.4602 / 0.6695; 5e-4 parent 0.4476 / 0.6248; 1e-3 0.4045 / 0.6009; 2e-3 0.2847 / 0.5457. The two cooldown forks are effectively tied on both all-range scores.

Full-data first epoch: 0.5551 / 0.6964. The GPU cooldown improves over its earlier 0.3862 / 0.5983 snapshot, but still trails the full-data reference. Exposure and schedules differ; the design set is small. All five evaluations use the same CUDA scoring recipe and cover 58000 stopped rollouts across 97 natural and 19 designed proteins; none needed extensions.

Historical checkpoint — October 27 update

The one-epoch v5p fork finished its four-update repair October 26 at 13:43 EDT. Final validation: 3.03582. Native step 186884 and all optimizer counters confirm exactly 186885 total updates; every exported parameter is finite. Tokenizer and model files are verified.

Completed full-data first epoch: 2.98441; continuation: 2.98099. Different exposure and schedules remain comparison limits. W&B summary is reconciled from durable final metrics; the invalid historical endpoint is preserved.

GPU epochs / target and latest loss: 5e-4 1.177 / 2, 3.08887; 1e-3 1.345 / 2, 3.11757; 2e-3 0.826 / 2, 3.57571; cooldown fork 0.866 / 1, 3.08435. All resumed and advanced.

2e-3 has repeated loss spikes (peak 4.09309). Keep the requested LR. At active compute speed, the GPU fork needs about 37 hours and the two-epoch runs 93–119 hours; interruptions add time. The October 21 R-precision snapshot is unchanged; no final TPU R-precision is available.

Historical recovery audit — October 26

GPU runs stopped October 22 following CoreWeave credential rotation. Reported success hid incomplete training. Last validation: 5e-4 3.09058; 1e-3 3.11904; 2e-3 3.15627; GPU fork 3.10000. Previous calendar ETAs are invalid.

TPU reached one epoch October 25, but the final padded update produced NaN parameters. Last finite scheduled validation: 3.03731 at step 186032. A tested padding-gradient fix preserves real-example gradients. Recovery restores step 186880 and replays exactly four updates; every exported parameter must pass a finite check.

TPU recovery a02 is queued for v5p-64 capacity. GPU parent a04 is starting after a real TensorStore cache read confirmed the credential fix. All four GPU recovery roots are submitted: 32 / 32 / 64 H100 for the LR sweep and 16 H100 for the GPU fork. New optimizer progress is not yet confirmed. R-precision results remain the completed October 21 snapshots below; no final TPU score is available.

ProteinMPNN-only learning-rate sweep

Experiment #364 asks whether the default 1.47B contacts-v1 model can train on redesigned sequences alone.

162,574,724 documents: 31,702,680 AFDB + 130,872,044 ESM-Atlas designs.

Controlled training recipe

Three learning rates: 0.0005, 0.001, 0.002. Two complete epochs, trained from scratch.

Preserve exp277 architecture, tokenizer, batch 128, context 8192, optimizer settings, WSD schedule fractions, validation and amino-acid order augmentation.

No native training documents. Fresh global shuffle each epoch.

Coverage and execution

23,921,189 packed examples per epoch; 373,769 updates for two passes.

Live regional audit passed: both source ledgers, raw tokens, packing and clipping exactly match the default corpus subset. Ten CPU tests, including the cooldown fork, passed.

The 64-H100 smoke passed ten updates with finite losses. Native checkpoint and HF weights plus tokenizer were verified.

All three production trials were launched on 32 H100s each, 96 total, using regional cached data. Saved W&B configurations match the default except for the requested changes.

The two-epoch runs are incomplete; see the October 26 recovery audit.

Validation comparison — 2026-10-21

Redesign-only training is behind the completed native-plus-redesign baseline at matched steps on the same full LM validation set.

At step 101,472: full data 3.07419; redesign-only LR 5e-4 / 1e-3 / 2e-3: 3.11629 / 3.13460 / 3.15598.

Latest LR 5e-4 validation: loss 3.09300 at step 181,804 versus full data 3.06074 (+0.03226 nats/token). LR 1e-3 is at 3.12220 at step 205,058. The lowest LR continues to lead this sweep.

Completed references and interpretation

Full-data first epoch: final loss 2.98441 at step 266,344. Additional full-corpus pass: 2.98099 at absolute step 479,417.

The continuation restores step 213,072 before cooldown; it is not an independent seed. Both completed trajectories include cooldown. The redesign-only two-epoch trials are still before cooldown; the TPU one-epoch fork entered cooldown on October 22.

The current runs also have longer warmup and amino-acid augmentation ramps. Equal optimizer steps match packed token slots, not actual tokens or proteins. The matched-step LM-validation gap and completed pre-cooldown contact evaluations both favor the full-data reference; the redesign-only trajectories are unfinished.

One-epoch cooldown fork — 2026-10-19

Fork the leading 5e-4 trajectory from full-state checkpoint 139,053. Preserve the original two-epoch run independently.

Keep the first-epoch shuffle, data offset, optimizer state, and original augmentation ramp. Retain LR 5e-4 until step 149,508, then linearly cool toward 5e-5 over the final 20% of one epoch.

Stop after 186,885 total updates (final checkpoint 186,884): exactly 23,921,189 packed examples, with 37 real examples in the final batch. The fork adds 47,831 updates to the inherited prefix.

At Oct 21, the fork has reached step 145,821: latest validation 3.10221 at 143,752 versus its parent's 3.10134 at the same step. Cooldown has not started yet. The parent remains independent. All ten CPU tests pass.

Repeated batch-priority preemptions have delayed progress, especially for the fork. Automatic checkpoint resumes are working, but calendar finish estimates remain sensitive to GPU availability.

Additional v5p cooldown copy: transfer verified

The independent TPU replica restores the same step-139053 full state and finishes at step 186884. Production requests v5p-64 (32 chips), us-east5-a, global batch 128 and batch priority. Existing GPU fork and parent are retained independently.

The approved 267.73 GB transfer finished at 15:59 UTC. All 21,936 objects passed independent inventory/checksum verification. The v5p-8 restore smoke passed ten updates (139054–139063) with finite losses and the expected accelerated LR decay. Final optimizer counters, both HF weight shards and tokenizer files were verified. Steady time: 27.09 s/update on four chips; its capped validation is not a full-data comparison.

Production was submitted at 16:33 UTC from commit 33cc077f and registered in W&B at 16:51:58 UTC. At 17:50 UTC it reached step 139702, with full validation loss 3.10391 and throughput about 244k token slots/s. Roughly 56 compute hours remain at that rate, before validation or interruptions. The source remains the original frozen checkpoint, not the smoke endpoint. Evidence: data/tpu_transfer_plan.json, tpu_input_verification.json, tpu_smoke.csv and tpu_smoke_verification.json.

Contact evaluation: approved continuation

Score 97 natural eval-val proteins and 19 eval-denovo designs. Eval-test is excluded. Latest complete checkpoints: 184314 (5e-4), 207865 (1e-3), 105722 (2e-3), 146264 (one-epoch fork). Exposure is unequal; cooldown has not started.

Keep the original 100-rollout, temperature-1, top-p-0.95 recipe. Start at 6L+128; the user approved continuing capped samples. Preserve each capped sample's exact prompt and generated tokens, then sample only its suffix within the remaining 8192-token context. Record continuation seeds, budgets and lengths. Require every sample to terminate; never discard capped samples from the score.

All four full evaluations completed: 46,400 rollouts, zero unfinished samples, and zero extensions needed. Thirty-seven CPU checks pass. Natural/design results and viral/low-MSA cuts stay separate.

Contact evaluation: durable sources restored

CoreWeave writes recovered after the account suspension. All four refreshed native sources are frozen and independently verified: 124 files, 70.63 GB. All four CPU exports passed verification, including both weight shards and tokenizer files. All four H100 evaluations succeeded by 22:47 UTC on October 21, without failures or preemptions. Model bytes stayed regional.

The earlier attempt lost temporary source objects during repeated preemptions. Its diagnostics and snapshot are preserved; no full evaluation groups completed. The fork smoke capped 2/100 samples on 5sbj_A, motivating the approved continuation.

Full-data first-epoch reference: eval-val 0.55510 / 0.53706; eval-denovo 0.69640 / 0.67497 (all / long). The continuation reference had eight capped samples on 5sbj_A. The new full evaluations needed no extensions. No matched redesign-corpus KNN null is available.

Contact results: natural proteins first

Eval-val, 97 natural proteins, all / long R-precision:

5e-4, step 184314: 0.39711 / 0.37791 1e-3, step 207865: 0.39562 / 0.37381 2e-3, step 105722: 0.38884 / 0.36465 GPU fork, step 146264: 0.38624 / 0.36627 Completed full-data default: 0.55510 / 0.53706

The redesign-only snapshots trail by 0.158–0.169 in all-range precision. The top two LR values are tied (difference 0.00148). Unequal exposure and unfinished cooldown prevent a final causal conclusion. Per-protein metrics, timings, bootstrap intervals and subgroup cuts are committed and public.

Contact results: 19 designs

Eval-denovo, all / long R-precision:

5e-4: 0.59671 / 0.54092 1e-3: 0.59087 / 0.53839 2e-3: 0.59242 / 0.53713 GPU fork: 0.59834 / 0.55408 Completed full-data default: 0.69640 / 0.67497

The small design set is a sanity check. All four snapshots precede cooldown, including the checkpoint labeled as the one-epoch fork. TPU weights are not part of this evaluation snapshot.

Historical training progress — October 22, 14:48 UTC

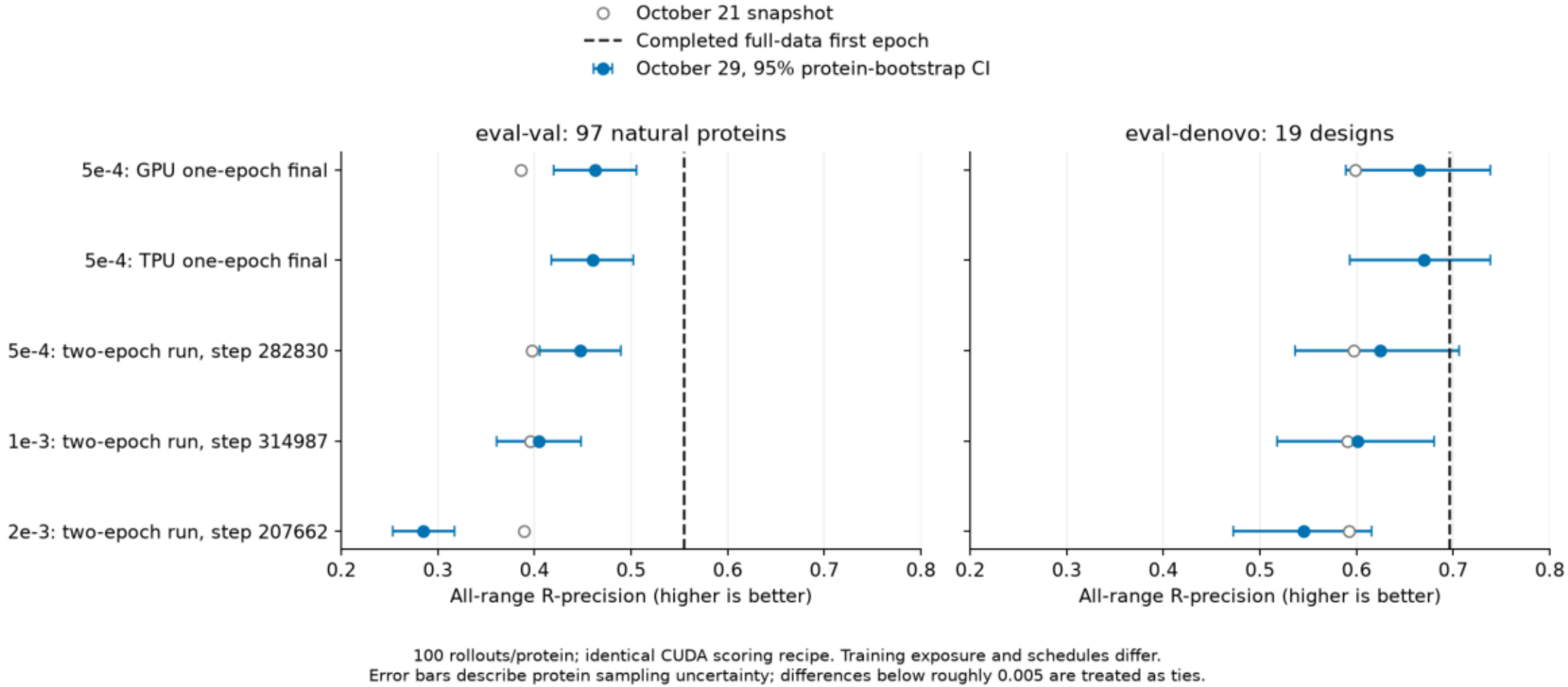
All five production runs are active. Epochs / target and latest validation loss:

5e-4 GPU: 1.072 / 2; loss 3.08945 1e-3 GPU: 1.186 / 2; loss 3.11922 2e-3 GPU: 0.618 / 2; loss 3.16114 GPU one-epoch fork: 0.791 / 1; loss 3.10318 v5p-64 one-epoch fork: 0.809 / 1; loss 3.10050

TPU cooldown has started. About 43 compute hours remain at current speed, putting the earliest finish on October 24; four prior preemptions show why calendar completion can be later. GPU fork: about 114 compute hours. Original two-epoch parent remains active and independent.

contact_r_precision.png

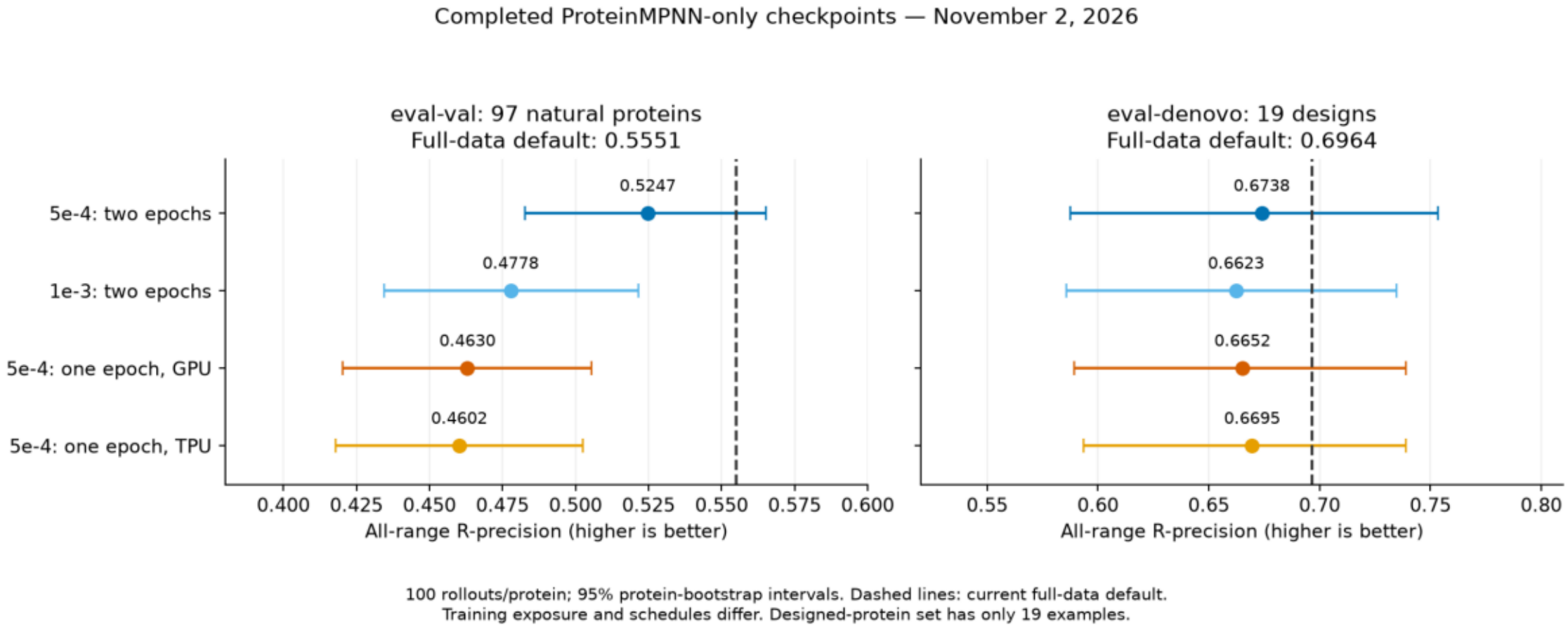
October 29 all-range R-precision with 95% protein-bootstrap intervals, October 21 snapshots, and the completed full-data first epoch. Natural proteins and designs are separate. Training exposure and schedules differ; all five evaluations use the same CUDA rollout recipe.




```
$ plot_contact_results.py
```

final_contact_r_precision.png

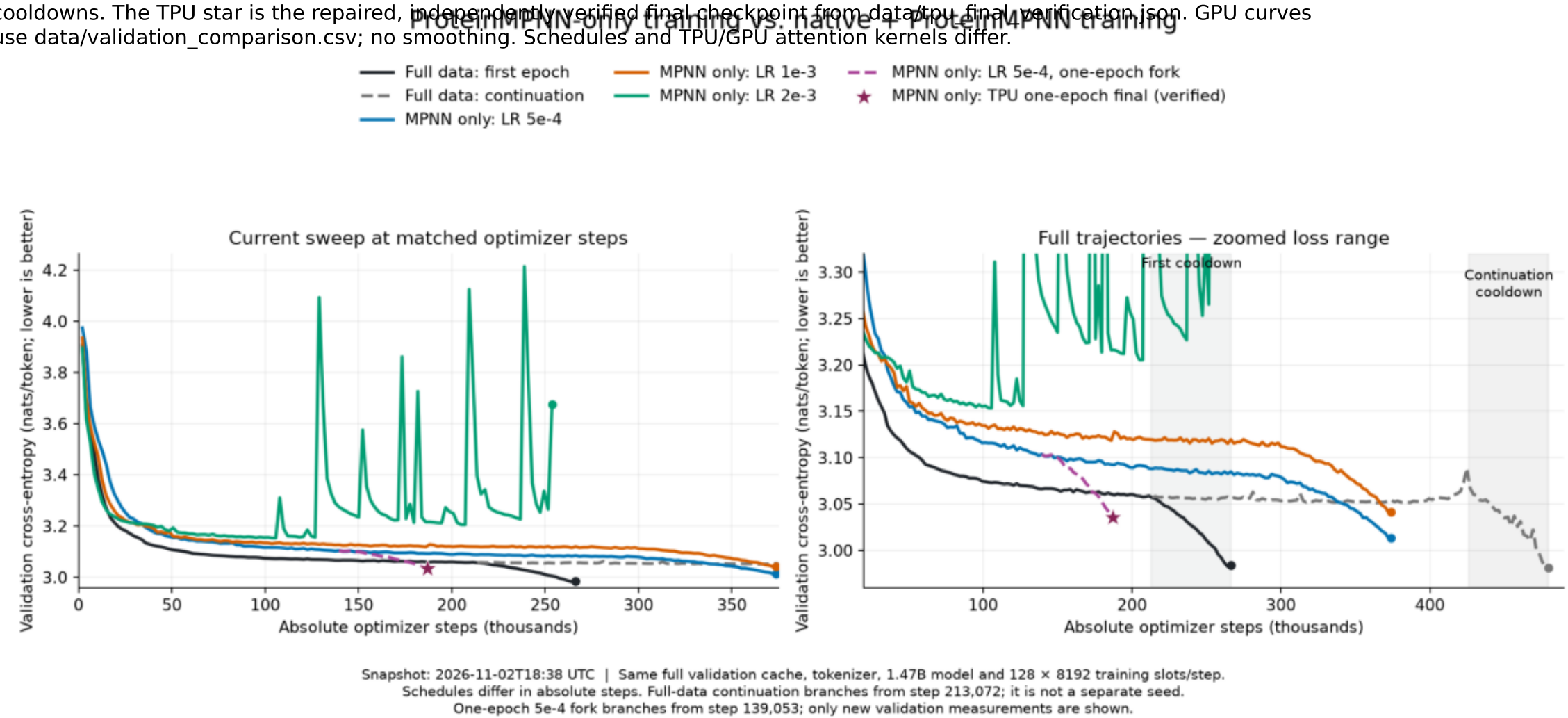
Final two-epoch step 373768 evaluations versus completed one-epoch forks and the exp277 full-data default. Macro all-range R-precision and 95% protein-bootstrap intervals; eval-val and eval-denovo are separate. The default uses its prior published evaluation.




```
$ plot_final_contact_results.py
```

validation_comparison.png

Exp364 validation against exp277 first epoch and its pre-cooldown continuation. Left: matched-step curves including the high-LR spikes. Right: full training with a zoomed loss range; spikes above the range are visible on the left. Shading marks completed-run cooldowns. The TPU star is the repaired, independently verified final checkpoint from data/tpu_final_verification.json. GPU curves use data/validation_comparison.csv; no smoothing. Schedules and TPU/GPU attention kernels differ.




```
$ compare_validation.py
```